

ImageNet-R-50 two-layer node-latent ablation

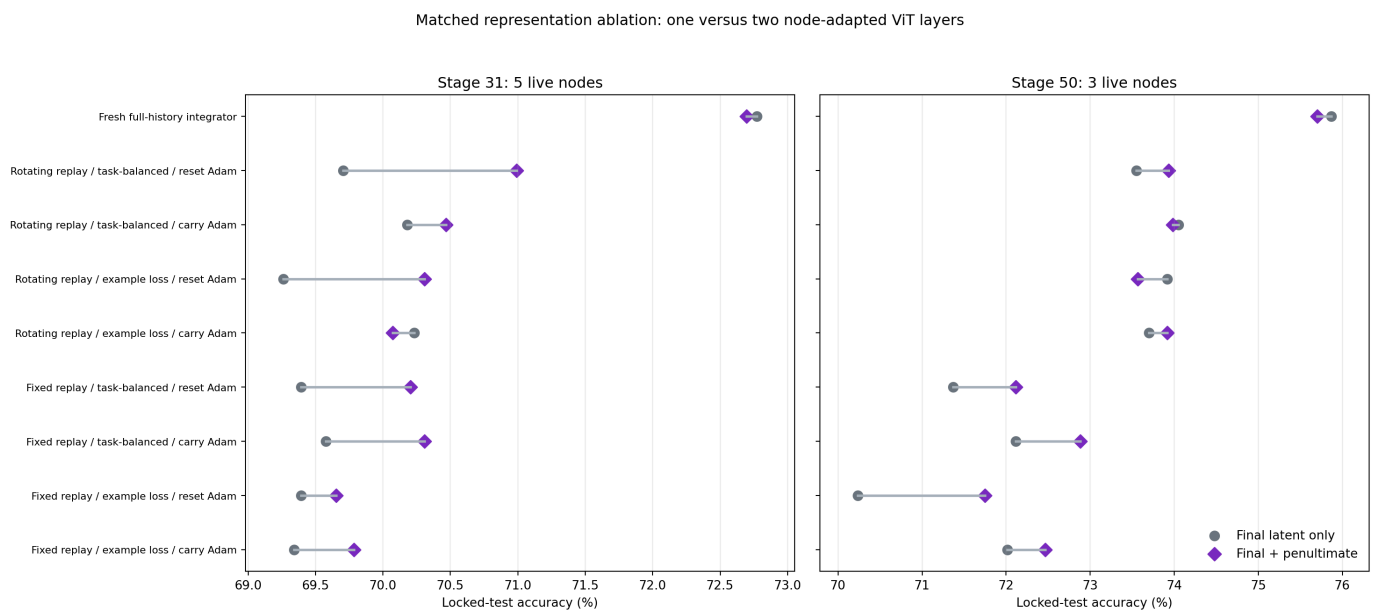
A nested representation expansion under the unchanged replay-adaptation matrix.

Finding. The validation-selected two-layer condition is **Rotating replay / task-balanced / carry Adam**; v6 selected **Rotating replay / example loss / carry Adam**. Across 16 matched online stage/condition cells, the added layer changes locked-test accuracy by +0.524 points. For the v6-selected condition, the stage-31 and stage-50 changes are -0.157 and +0.217 points.

Interpretation. Fresh full-history test accuracy declines slightly at both checkpoints, so the extra token does not raise the observed integration ceiling. The substantially larger validation gain is consistent with the fact that upstream nodes saw those validation identities during fitting.

Headline comparison

Stage	Single-layer selected	Two-layer selected	Single-layer full history	Two-layer full history	True-node oracle	Joint IID
31	70.231	70.466	72.773	72.694	83.700	80.294
50	73.700	73.983	75.867	75.700	81.117	78.867



Each line joins the same replay, weighting, optimizer, stage, and test identities.

Exact representation change

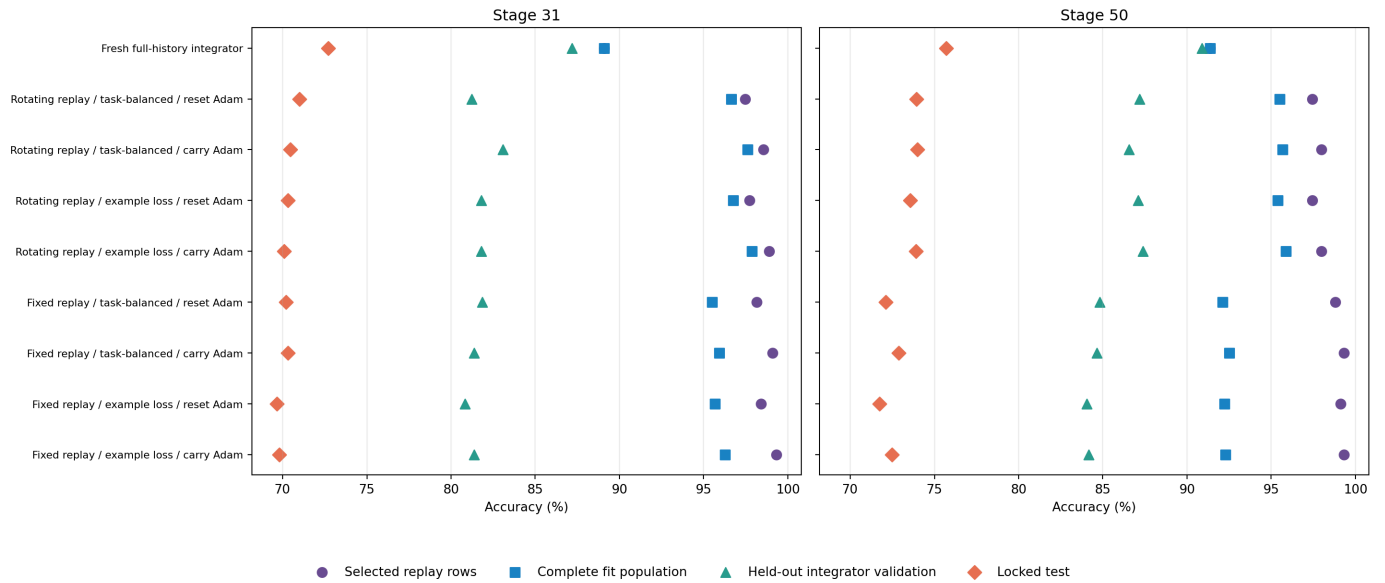
The new 768-value class token is captured after the penultimate transformer block while the live node's own LoRA is installed. It is appended after the unchanged 1,369-value slot prefix and normalized separately. The MLP grows from 9,122,760 to 13,841,352 parameters. New input columns start at zero; all compatible input columns and downstream parameters copy the v6 initialization.

Stage	Condition	Single val	Two-layer val	Val delta	Single test	Two-layer test	Test delta
31	Fresh full-history integrator	86.979	87.176	0.197	72.773	72.694	-0.079
31	Rotating replay / example loss / carry Adam	81.076	81.797	0.722	70.231	70.073	-0.157
31	Rotating replay / example loss / reset Adam	78.518	81.797	3.280	69.261	70.309	1.048
31	Rotating replay / task-balanced / carry Adam	80.485	83.076	2.591	70.178	70.466	0.288
31	Rotating replay / task-balanced / reset Adam	79.829	81.240	1.410	69.706	70.991	1.284
31	Fixed replay / example loss / carry Adam	79.731	81.371	1.640	69.340	69.785	0.445
31	Fixed replay / example loss / reset Adam	79.895	80.813	0.918	69.392	69.654	0.262
31	Fixed replay / task-balanced / carry Adam	79.731	81.371	1.640	69.575	70.309	0.734
31	Fixed replay / task-balanced / reset Adam	78.911	81.863	2.952	69.392	70.204	0.812
50	Fresh full-history integrator	91.167	90.896	-0.271	75.867	75.700	-0.167
50	Rotating replay / example loss / carry Adam	85.917	87.375	1.458	73.700	73.917	0.217
50	Rotating replay / example loss / reset Adam	85.688	87.083	1.396	73.917	73.567	-0.350
50	Rotating replay / task-balanced / carry Adam	85.979	86.542	0.562	74.050	73.983	-0.067
50	Rotating replay / task-balanced / reset Adam	85.854	87.187	1.333	73.550	73.933	0.383
50	Fixed replay / example loss / carry Adam	83.229	84.167	0.937	72.017	72.467	0.450
50	Fixed replay / example loss / reset Adam	80.896	84.042	3.146	70.233	71.750	1.517
50	Fixed replay / task-balanced / carry Adam	83.479	84.646	1.167	72.117	72.883	0.767
50	Fixed replay / task-balanced / reset Adam	83.083	84.812	1.729	71.367	72.117	0.750

Two-layer adaptation and generalization

Stage	Condition	Selected train	Full fit	Validation	Test	Old-task test	Current-task test
31	Fresh full-history integrator	89.101	89.101	87.176	72.694	72.196	72.807
31	Rotating replay / example loss / carry Adam	98.890	97.868	81.797	70.073	69.521	71.930
31	Rotating replay / example loss / reset Adam	97.710	96.744	81.797	70.309	70.164	68.421
31	Rotating replay / task-balanced / carry Adam	98.551	97.605	83.076	70.466	70.059	70.175
31	Rotating replay / task-balanced / reset Adam	97.476	96.638	81.240	70.991	70.552	71.930
31	Fixed replay / example loss / carry Adam	99.311	96.269	81.371	69.785	69.062	74.561
31	Fixed replay / example loss / reset Adam	98.399	95.662	80.813	69.654	69.186	71.053
31	Fixed replay / task-balanced / carry Adam	99.089	95.941	81.371	70.309	69.948	70.175
31	Fixed replay / task-balanced / reset Adam	98.166	95.490	81.863	70.204	69.831	71.053
50	Fresh full-history integrator	91.359	91.359	90.896	75.700	75.003	80.909
50	Rotating replay / example loss / carry Adam	97.976	95.875	87.375	73.917	73.418	77.273
50	Rotating replay / example loss / reset Adam	97.449	95.385	87.083	73.567	72.943	80.000
50	Rotating replay / task-balanced / carry Adam	97.988	95.667	86.542	73.983	73.538	79.091
50	Rotating replay / task-balanced / reset Adam	97.449	95.516	87.187	73.933	73.487	79.091
50	Fixed replay / example loss / carry Adam	99.321	92.276	84.167	72.467	71.717	80.000
50	Fixed replay / example loss / reset Adam	99.123	92.224	84.042	71.750	70.911	80.909
50	Fixed replay / task-balanced / carry Adam	99.310	92.510	84.646	72.883	72.353	80.000
50	Fixed replay / task-balanced / reset Adam	98.807	92.120	84.812	72.117	71.451	78.182

Memorization check: selected rows versus unseen populations

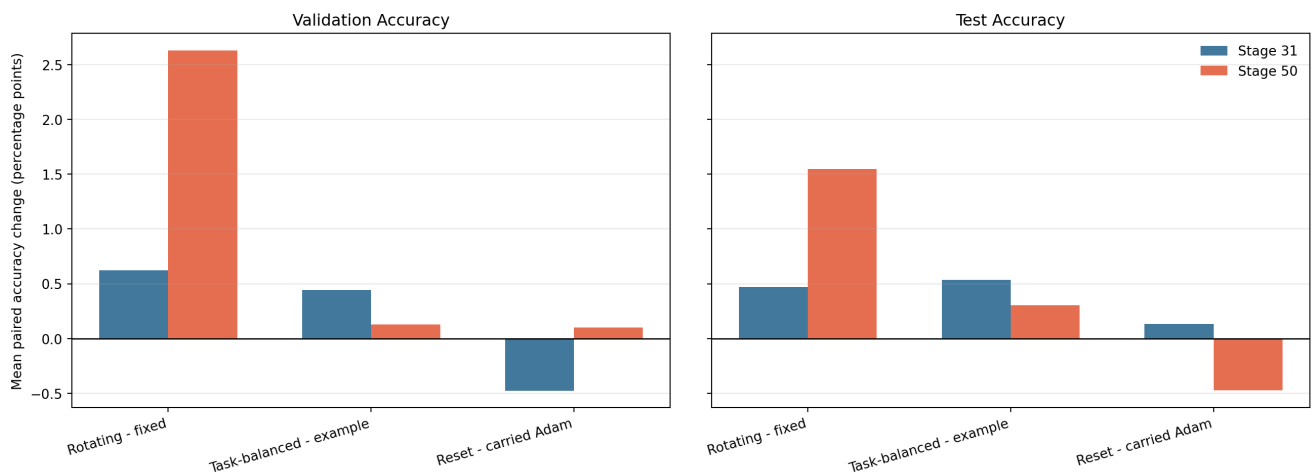


The full-fit and unseen-population gaps measure whether repeated online fitting adapts beyond retained identities.

Within-run replay-factor effects

Factor	Metric	Mean paired change (pp)
reset_minus_carried_adam	test_accuracy	-0.170
reset_minus_carried_adam	validation_accuracy	-0.188
rotating_minus_fixed	test_accuracy	1.009
rotating_minus_fixed	validation_accuracy	1.627
task_balanced_minus_example	test_accuracy	0.421
task_balanced_minus_example	validation_accuracy	0.286

Main effects averaged over the other two paired factors



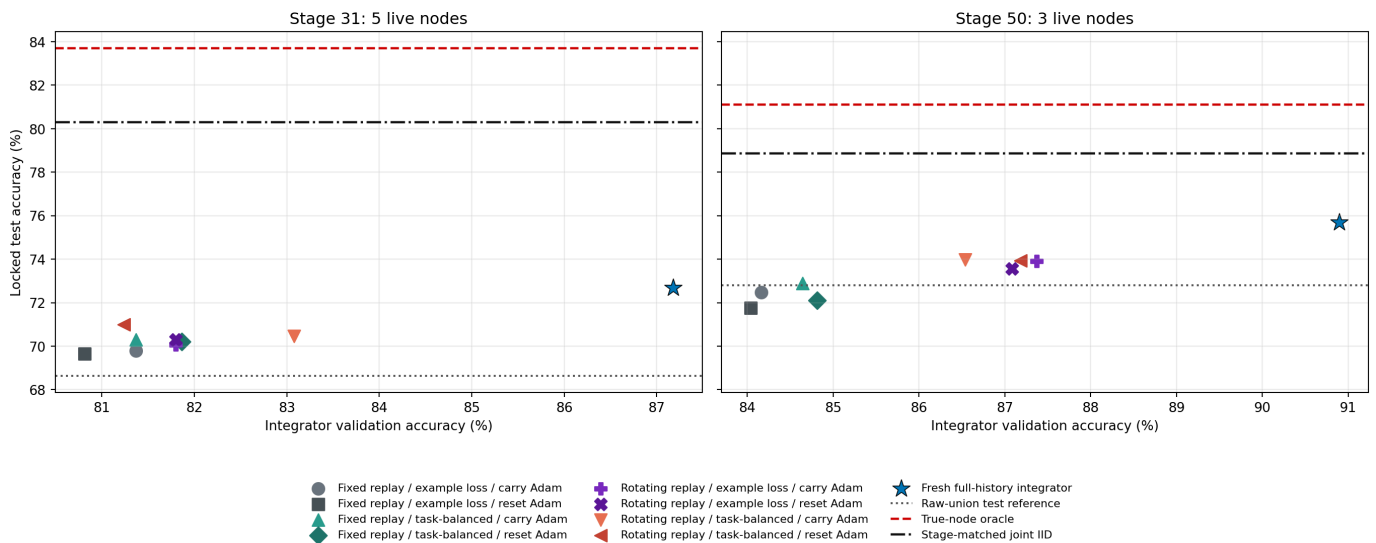
Interpretation boundaries

This is one seed and a test set examined by earlier experiments. Full-history retrains only the integrator and is not online. Integrator validation identities were held out from integrator fitting but not from upstream LoRA-node fitting.

Reproducibility and work

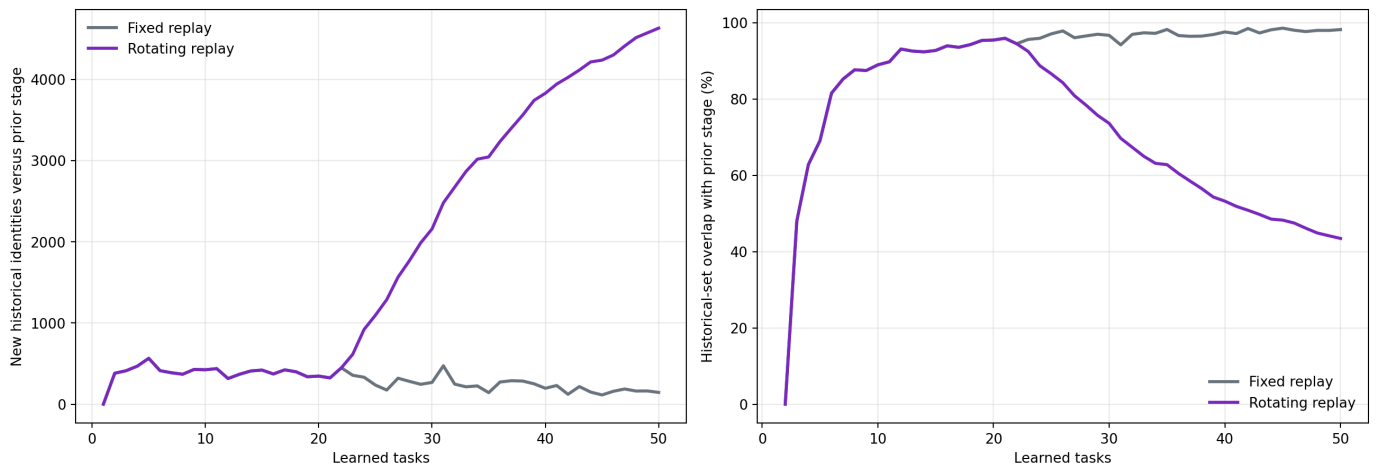
Condition	Parameters	Image presentations	Optimizer steps	Fit seconds
Fixed replay / example loss / carry Adam	13841352	1375876	2760	21.855
Fixed replay / example loss / reset Adam	13841352	1375876	2760	21.428
Fixed replay / task-balanced / carry Adam	13841352	1375876	2760	21.767
Fixed replay / task-balanced / reset Adam	13841352	1375876	2760	21.998
Rotating replay / example loss / carry Adam	13841352	1375876	2760	21.535
Rotating replay / example loss / reset Adam	13841352	1375876	2760	20.596
Rotating replay / task-balanced / carry Adam	13841352	1375876	2760	20.625
Rotating replay / task-balanced / reset Adam	13841352	1375876	2760	20.746
Fresh full-history integrator	13841352	1895834	3744	71.430

Replay adaptation: validation-selected evidence and locked test accuracy

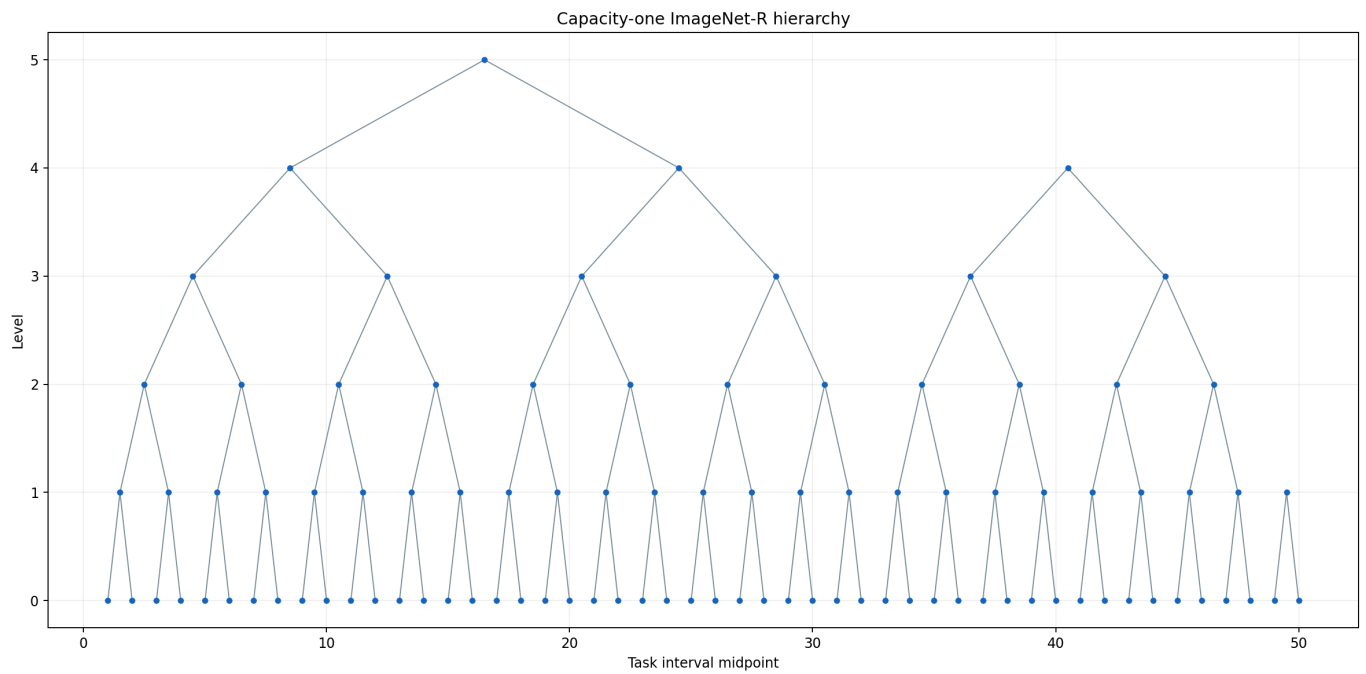


All two-layer online conditions with identical stage-matched references.

Stage-keyed replay changes which old examples train the integrator



Replay membership is unchanged from v6.



The reused source hierarchy has five live nodes at stage 31 and three at stage 50.

Report source SHA-256: f339f7af7fad9f60e1f895e5d2fe55183ba354e7cc21dfbfe7aaa69921a7b510